AKASH SARAVANAN

GEOSPATIAL ENGINEER | GIS · WEB GIS · GEOAI

Chennai, India · Open to relocation (Bengaluru) | +91 79040 81911 | akashcivil04@gmail.com

Akash/LinkedIn | Akash/Portfolio | Akash/GitHub

PROFILE SUMMARY

Geospatial Engineer with nearly four years of experience delivering GIS mapping, remote sensing, spatial analysis, and geodatabase management across infrastructure and land-development projects. Specialised in QGIS, ArcGIS Pro/Online, Google Earth Engine, and Python-based geospatial automation (GDAL, Rasterio, GeoPandas, PyQGIS). Leverage AI tools (Claude, ChatGPT) and AI-assisted development to build Web GIS platforms end to end and to automate day-to-day GIS workflows, and apply GeoAI and machine-learning models for land-use analysis and automated feature extraction. Led QA/QC processes achieving 100% data integrity across cadastral and property-tax geospatial projects while mentoring a two-person team. Focused on translating complex spatial data into accurate, decision-ready outputs that support infrastructure planning and land-development decisions.

Highlights: 30+ CMDA layout projects on a Web GIS platform I built · 6,000+ cadastral parcels georeferenced or stitched and 5,000+ acres mapped · 25% QC workload cut · Led property-tax QA/QC for 60 wards (up to 5,000 buildings each) at 100% data integrity · 13 of 15 parcels auto-placed within 5 m

TECHNICAL SKILLS

- GIS & Mapping Tools: ArcGIS Pro, ArcGIS Online, Field Maps, QGIS, Google Earth Engine, AutoCAD Map 3D, Google Earth Pro, Global Mapper
- Remote Sensing & Earth Observation: Google Earth Engine, satellite imagery analysis, vegetation indices (NDVI), image classification, GeoAI / machine learning for feature extraction and land-use prediction
- Data & Geodatabase Management: Geodatabase structuring, metadata documentation, folder structure/naming conventions, version control, vector/raster conversion, georeferencing, geoprocessing
- BIM – GIS Integration: Coordinating GIS/Spatial data with CAD & BIM drawings, leveraging civil engineering domain knowledge.
- Development & Automation: Python (GDAL, Rasterio, GeoPandas, Shapely, PyQGIS — geospatial workflow automation), PostGIS, SQL / GeoPackage, web scraping & REST APIs (Requests, Selenium), geocoding (Nominatim); JavaScript (React, Leaflet, MapLibre GL) for Web GIS with AI-assisted development; LLM APIs (Gemini) for image-based verification
- Drone Data Processing: Pix4Dmapper, Agisoft Metashape, RealityCapture, DJI Terra
- Spatial Analysis & Cartography: Multi-criteria spatial analysis, spatial data analysis, cartographic map production, symbology and layout design

EXPERIENCE

FARMWISEAI PRIVATE LIMITED – Geospatial Engineer; Chennai, India *September 2025 – Present*

- Techno-Economic Feasibility Studies (TEFR): Conducting spatial and economic analysis for proposed industrial parks — land potential, infrastructure connectivity and investment viability (recent: Perungalathur 476 ac, Kattupakkam 574 ac) — and coordinating GIS datasets with CAD/BIM survey drawings so layouts stay consistent across disciplines.
- Site Suitability Modeling: Developed multi-criteria decision-making models to identify optimal locations for industrial setups, balancing technical constraints with environmental and logistical factors.
- Precision Drafting & Layout: Executing end-to-end AutoCAD FMB (Field Map Book) stitching, transforming raw survey data into complete, high-accuracy industrial layouts — 700+ parcels stitched and 5,000+ acres mapped to date; georeferenced ~5,370 cadastral parcels across 19 villages for a 32 km road corridor, delivered topology-clean.
- Quality Assurance (QA/QC): Leading a two-person QC team in QGIS that has verified property-tax data for 60 municipal wards, up to 5,000 buildings each, against the tax database at 100% data integrity.
- Web GIS Platform Development: Built the CUMTA (Chennai Unified Metropolitan Transport Authority)

Web GIS platform end to end with AI-assisted development (Claude, ChatGPT): React/Leaflet viewer, FastAPI backend with role-based access, and a DWG/DXF-to-GIS pipeline serving 30+ CMDA layout projects.

- FPO Portal Contribution: Contributing frontend UI improvements and building the land-document retrieval workflows (Patta/Chitta, A-Register) that feed accurate cadastral data into an FPO (Farmer Producer Organization) platform.
- GIS Automation & GeoAI: Streamlining daily GIS workflows with Python scripts and tools that cut manual data entry, and applying GeoAI / machine-learning models to spatial data for predictive land-use analysis and automated feature extraction.
- Geospatial Data Packages: Assembling state- and country-level GIS packages for due diligence (Maharashtra CRZ/MIDC and 610k+ coastal plots, Kenya farmer-services data, Andhra Pradesh cadastral services) as QGIS projects on GeoPackage in the right projected CRS.
- Stakeholder Presentation & Field Data: Turning suitability reports into visual, data-driven presentations for executive decisions; ArcGIS Online dashboards and Field Maps for site assessment and survey capture.

OPERATIONAL ENERGY GROUP INDIA LIMITED – Chennai, India *January 2023 – August 2025*

I. OEG DEFENCE AND AEROSPACE PRIVATE LIMITED – GIS Analyst; *Jun 2024 – Aug 2025*

- Developed GIS datasets and topographic maps for engineering applications.
- Produced cartographic outputs (site maps, layout plans, thematic maps) using QGIS for engineering review and client submission.
- Used ArcGIS Online and Field Maps for field data collection and visualization to support project data workflows.
- Ensured positional accuracy and consistency of map outputs through systematic QA/QC checks.

II. OPERATIONAL ENERGY GROUP INDIA LIMITED – GIS Analyst; *Jan 2024 – May 2024*

- Conducted volumetric analysis and topographic mapping for thermal power plant sites.
- Prepared land survey-based mapping deliverables to support infrastructure planning.
- Maintained mapping standards and symbology conventions across project deliverables.

III. ZUPPA OEG GEN5 TECHNOLOGIES PRIVATE LIMITED – GIS Analyst; *Jan 2023 – Dec 2023*

- Processed drone-captured imagery into georeferenced Orthomosaics and digital elevation models for site mapping.
- Generated topographic outputs from UAV survey data for engineering and client review.
- Supported geospatial data integration between drone-derived datasets and GIS platform (QGIS) for mapping deliverables.
- Provided mapping-related technical support and demonstration at client meetings.

SSG ENGINEERS – Site Engineer; Salem, India *December 2021 – December 2022*

- Monitored daily construction activities to ensure adherence to project plans and safety standards.
- Coordinated with contractors and suppliers to maintain efficient workflows and timely material deliveries.
- Maintained accurate documentation of project progress inspections and site reports.

EDUCATION

PANIMALAR ENGINEERING COLLEGE – Chennai, INDIA *September 2020*
Bachelor of Engineering, Civil Engineering; Cumulative GPA: 7.32/10.0

THIAGARAJAR POLYTECHNIC COLLEGE – Salem, INDIA *May 2017*
Diploma, Civil Engineering; Cumulative Percentage: 86.74

PROJECTS

- **CUMTA Web GIS Platform (PoC):** Full-stack Web GIS platform for the Chennai Unified Metropolitan Transport Authority managing CMDA layout data across 30+ layout projects, built with AI-assisted development (Claude, ChatGPT): React/Leaflet viewer, FastAPI backend with JWT role-based access, a CAD ingestion pipeline turning DWG/DXF layouts into georeferenced GeoJSON with GCP correction, a road register with connectivity and width-compliance checks, Shapefile/KML/Excel/PDF exports and a SQLite/PostGIS data layer, backed by 250+ automated tests.
- **FPO Portal:** Map-first management cockpit for Farmer Producer Organizations, where a survey-level land map and a live MIS dashboard work as one instrument. Integrated real government data (land parcels, soil-health cards, fertilizer dealer stock) and built the village-drill and parcel-selection experiences on MapLibre GL with statewide PMTiles layers for a 323-member FPO in Tamil Nadu.
- **Automated FMB Georeferencing, Tambaram–Chengalpattu Rail Corridor (R&D, open source):** Python pipeline that converts Tamil Nadu Field Measurement Book (FMB) PDFs into vector parcels with every printed dimension preserved, then places each sheet by matching shared boundaries with already-placed neighbours (rotation and shift only, scale fixed), with green/amber/red confidence flags for review. In leave-one-out validation on a calibration village, 13 of 15 parcels were re-placed within 5 m. Built with GeoPandas, Shapely, Rasterio, OpenCV and SciPy, 120+ unit tests; now being packaged as a QGIS plugin.
- **Site Suitability Desktop App for SIPCOT Sites (open source):** Desktop app that turns a site boundary into the full preliminary site analysis — survey parcels, CLIP subdivisions, A-register extents, guideline values, crop data, land cost, connectivity maps and a 14-slide deck — with portal answers cached in PostGIS.
- **Property-Tax Street Audit, Coimbatore City Municipal Corporation (Proof of Concept, open source):** Python tool that checks property-tax building records against street-level imagery: it picks panoramas with a real line of sight to each footprint, has a vision LLM (Gemini) read floors, usage and shop fronts blind, and lets fixed rules decide the verdict against the survey and tax record.
- **PyQGIS QC Automation, Coimbatore City Municipal Corporation:** Built a PyQGIS tool that flags spatial mismatches between building-footprint and geocode shapefiles across the citywide property-tax dataset (residential, commercial and industrial), with custom in-QGIS shortcuts that take reviewers straight to each flagged feature. Reduced manual quality-control workload by 25%.
- **Site Suitability & Techno-Economic Feasibility (TEFR) for Proposed Industrial Parks:** Multi-criteria spatial analysis combining vector (land use, infrastructure proximity) and raster (terrain, environment) datasets into suitability maps for site selection, plus GIS layouts and location intelligence for techno-economic feasibility reports informing planning and investment viability.
- **Cadastral FMB Stitching & Large-Scale Land Parcel Mapping:** AutoCAD-based FMB stitching across sites of 200 to 2,000 acres — 700+ parcels stitched, 5,000+ acres mapped — with boundary integrity for cadastral records used in land acquisition and legal documentation.
- **Photogrammetry Mapping & Volumetric Estimation (Highways, Railways & River Beds):** Processed aerial survey data into DSMs, orthomosaics and contour maps for National Highways, railway corridors and river beds, and estimated coal-stockpile volumes from point-cloud and elevation data.

KEY COMPETENCIES

- GIS · Remote Sensing · Spatial Analysis · Photogrammetry · Cadastral Mapping · Geodatabase Management · Data QA/QC
- Web GIS · GeoAI / Machine Learning · Python · JavaScript · PostGIS · Data Analysis · Project Planning & Delivery · Stakeholder Communication

CERTIFICATIONS AND TRAINING

- AutoCAD Certification (2017) · AI/ML for Geodata Analysis — IIRS, ISRO (2024) · Essentials for PyQGIS: Python for Geospatial Automation — Udemy (2025) · Drone Photogrammetric Image Processing — Udemy (2024) · GIS Basics — Esri (2023) · ArcGIS Pro Essential Training — LinkedIn Learning (2023)